

MolmoAct2 · arXiv 2605.02881 · Ai2 / allenai

Stakeholder

18+3 · all seats empty

Open ARM vs π opacity

LIBERO 97.2 / Think 98.1 · Spaces 37.7 · DROID 87.1

Oct 28 paper club · VLA Architectures · quiet build

Spine: geometric S2 $\neq$ semantic HL $\neq$ stage voting · FM motor

FALSE: “has $\pi_{0.5}$ HL” / “has RTC” / “has BEHAVIOR S2 voting”

Open Action Reasoning Model (ARM): geometric System-2

(adaptive depth tokens in Think) before a per-layer-KV DiT FM action expert

— **deployable open alternative to closed π -series.**

Backbone

Molmo2-ER
4B + SigLIP2

Curriculum

FAST AR →
FM expert (KI) → FT

Expert

~621M DiT
L=36 · N=10 Euler

Think

100 depth codes
adaptive reuse

Claims tattoo: LIBERO 97.2 / Think 98.1 ($\pi_{0.5}$ 96.9) · Spaces 37.7 vs 34.5 · DROID OOD 87.1%

Rates: ~55.8 Hz Act2 / ~12.7 Hz Think @ H100 H=10 + CUDA Graphs · RTC absent

Open ARM vs π opacity

MolmoAct2 (open)

- ✓ Weights + code + data claimed
- ✓ 720 h BimanualYAM public
- ✓ OOTB SO-100 / Franka / YAM
- ✓ Think geometric S2 open
- ✓ allenai/molmoact2

π -series (closed)

- Frontier weights gated
- Proprietary data walls
- HL/RTC stack opaque to outsiders
- Strong but not freely FT-able
- Baseline via OpenPI subsets

Program decision

Use as open FM-VLA + ARM baseline when π gated.
Competitive on paper protocols: LIBERO 97.2/98.1 · Spaces 37.7 · DROID 87.1%.
Budget H100 FT (~1-2k GPU-h/embodiment) + camera calibration discipline.

KPI tattoo — must-quote numbers

LIBERO avg

97.2%

Think 98.1%
 $\pi_{0.5}$ 96.9%

LIBERO Long

93.2→95.4

Think headroom
(ceilings elsewhere)

MolmoSpaces

37.7

vs $\pi_{0.5}$ 34.5
(+3.2)

DROID OOD

87.1%

vs $\pi_{0.5}$ 45.2
real 15-trial

SO-100

56.7%

vs π_0 45.3

YAM 8-task

~50%

vs OFT 35.5

ER VLM

63.8%

9/13 best

Hz Act2/Think

55.8 / 12.7

H100 Graphs

CRITICAL SPINE — three outer loops (do NOT conflate)

All three call an FM chunk expert (System 1 / motor). Different intermediate discrete states.

MolmoAct2 Think

Adaptive depth / geometry

- 10×10 depth codes (100)
- VQ depth lattice
- Regen if RGB cos < 0.996
- Cached static cells
- Geometric System-2

NOT semantic subtasks

$\pi_{0.5}$ HL

Semantic subtask language

- AR subtask strings
- Optional boxes
- Task language hierarchy
- Then FM LL chunk
- Semantic System-2

NOT depth lattice

BEHAVIOR'25 S2

Stage IDs + voting

- Task-progress stages
- Classifier + voting
- Non-Markov memory
- Then FM expert
- Checklist System-2

NOT scene geometry

FALSE claims: “has $\pi_{0.5}$ HL” · “has RTC” · “has BEHAVIOR S2 voting”

TENTATIVE: BEHAVIOR may want depth + stage complementary · RTC absent — latency via FM cache + CUDA Graphs + Think reuse

L0

Motion / Perception + language **control hierarchy — L0 → L1 → L2**

SigLIP2 + LLM multi-cam

Interface: tokens + per-layer K/V → expert



L1

System-2 (optional Think)

AR depth codes (10×10)

Interface: depth → VLM context → expert KV



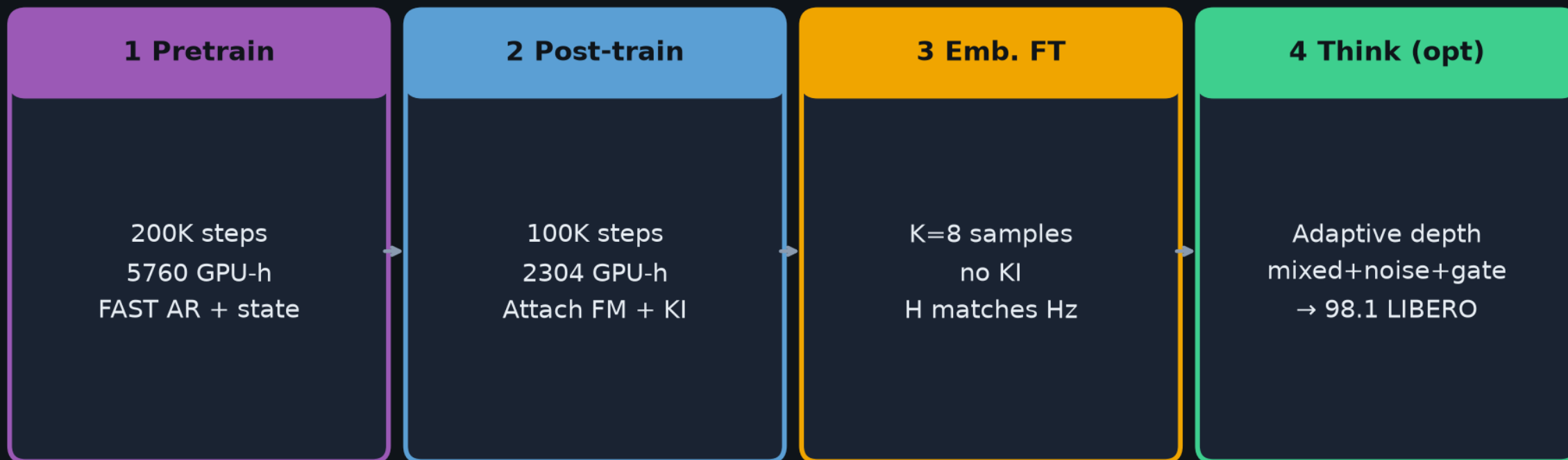
L2

Motor

DiT FM · $H \approx H_z$ · $N=10$ Euler

Interface: continuous joints / delta-EE

Curriculum — FAST discrete → FM continuous



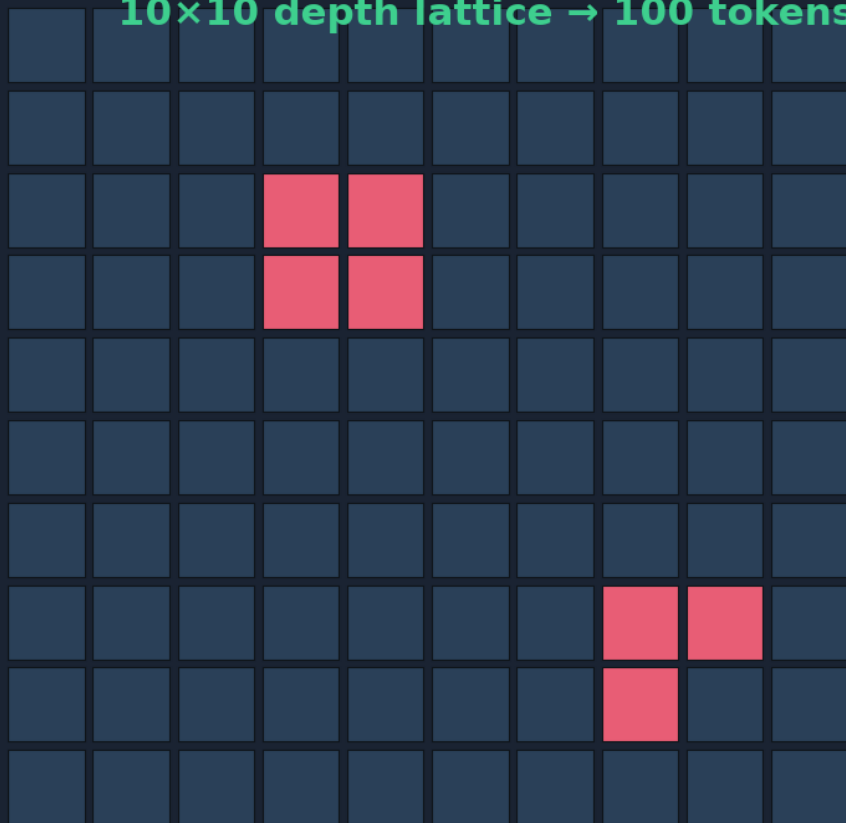
Deploy default = continuous FM path (not discrete AR actions)

Expert $\sim 621\text{M}$ · L=36 · AdaRMS on t · N=10 Euler · $H \in \{10, 15, 30\}$ · $D \leq 32$

MolmoAct2-Think — adaptive depth codes (geometric S2)

NOT $\pi_{0.5}$ semantic HL · NOT BEHAVIOR stage IDs

10×10 depth lattice → 100 tokens



pink = regenerate (RGB cos < 0.996) · blue-gray = cache reuse

1. Observe RGB (+ multi-cam)

2. AR depth codes <depth_0...127>

3. Reuse static cells / regen changed

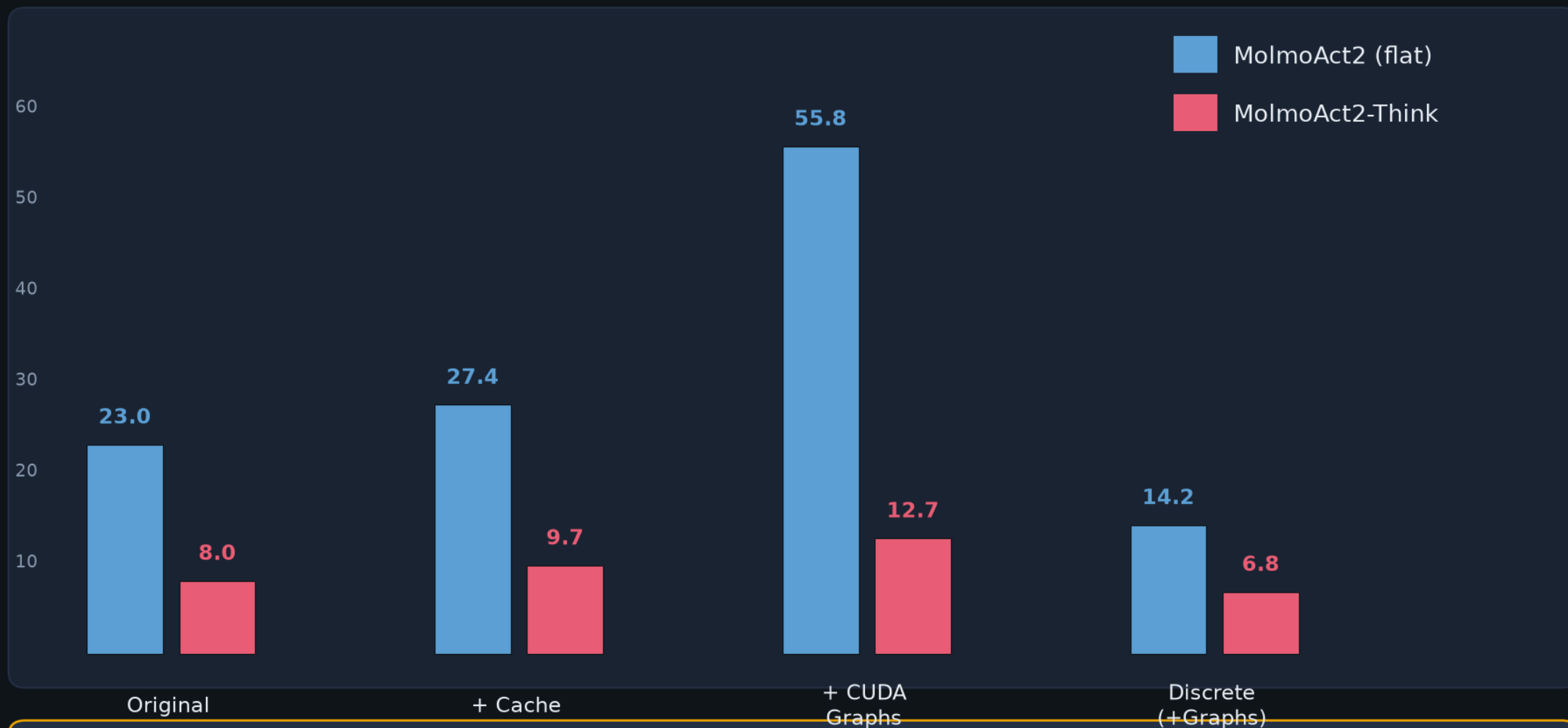
4. Optional per-layer depth gate

5. Gate into expert KV → FM chunk

Latency tax: 12.7 Hz Think vs 55.8 Hz flat (H100, H=10, Graphs)

Inference control rate — Think on/off + optimizations

$H=10 \cdot 1 \times H_{100} \cdot \text{LIBERO} \cdot \text{Hz} = H / \text{end-to-end chunk latency}$



RTC absent — latency via FM cache + CUDA Graphs + Think reuse · don't quote SR without Hz

Fig.1 · MolmoAct2 overview



Figure 1 Overview of MolmoAct2. MOLMOACT2 is a fully open action reasoning model for real-world deployment. From a suite of high-quality robot datasets that we collect, filter, and curate at scale across three platforms spanning the low-to-medium cost range (**left**), we train MOLMOACT2 and its adaptive-depth reasoning variant MOLMOACT2-THINK, coupled to the VLM backbone through per-layer KV conditioning (**center**). The resulting models deploy out-of-the-box on bimanual YAM, SO-100/101, and DROID Franka, and adapt to in-the-wild tasks such as cleaning up, washing dishes, wetlab automation, and pouring tea (**right**).

1 Introduction

Physical intelligence is fundamentally organized around perception and action (Tversky, 2025). Rather than reasoning over abstract internal computation, human think by constructing spatial representations, simulating actions, and interacting with the world through their bodies (Tversky, 2019, 2013). Although we conflate today's robot foundation models as displaying such intelligence; from a cognitive science perspective, they remain incomplete models of intelligence. They often lack structured spatial representations Qu et al. (2025), rely on heavyweight internal reasoning processes that impede real-time interaction (Lee et al., 2025; Kim et al., 2026a; Zhu et al., 2025), and are difficult to adapt or extend due to limited openness (Black et al., 2024; Intelligences et al., 2025).

Fig.2 · training data mixture

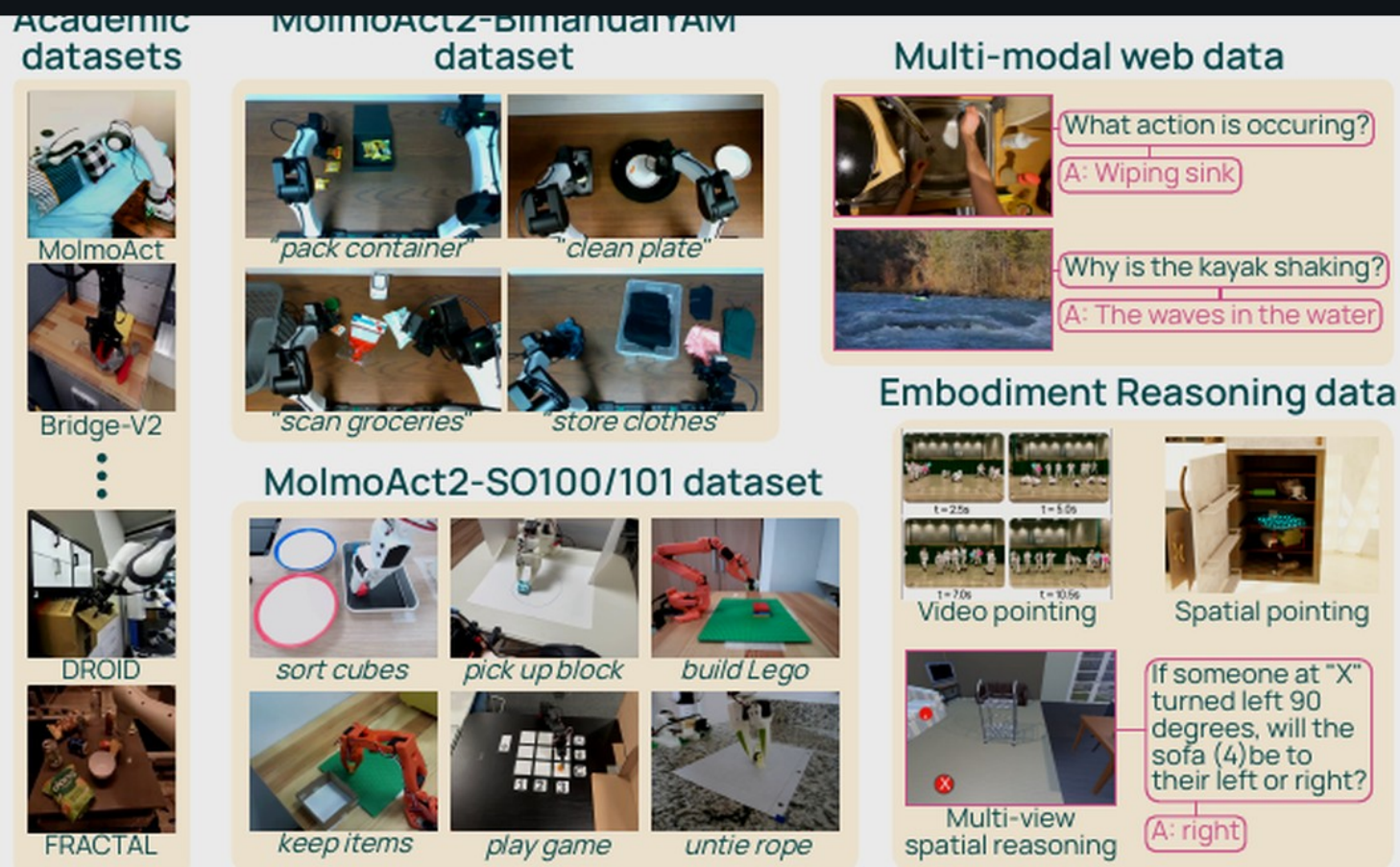


Figure 2 Overview of the MolmoAct2 training data. Our data mixture combines public academic robot datasets, MOLMOACT2-BIMANUALYAM DATASET, MOLMOACT2-DROID DATASET, MOLMOACT2-SO100/101 DATASET, multimodal web data, and embodied-reasoning data.

checks (required schema fields, valid action/state tensors, no NaN/corrupt samples), (ii) removal of eval-style datasets, (iii) license/codebase eligibility checks, and (iv) a final TOPReward quality gate (Chen et al., 2026). In stage (iv), we keep datasets whose mean TOPReward over the last 3 sampled episodes is above a threshold obtained by averaging the TOPReward over a collection of human-audited high-quality datasets.

The filtered data spans both SO-100 and SO-101 embodiments, multiple camera configurations, varied object

Fig.3 · BimanualYAM hardware

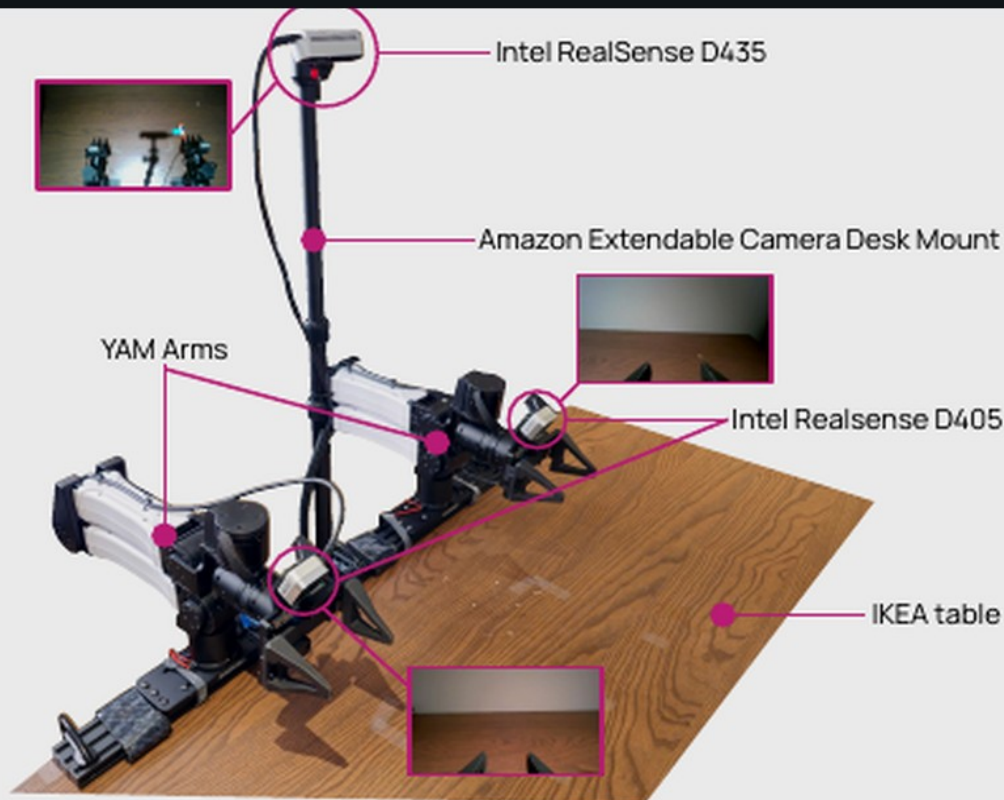


Figure 3 MolmoAct2-BimanualYAM Dataset collection setup. Our standardized MOLMOACT2-BIMANUALYAM DATASET collection setup. Every component is readily available for purchase, and the total cost of the entire setup is under \$6,000 USD.

3.4 Language annotation pipeline

Following standard practice in imitation learning, each demonstration in our datasets is paired with a language instruction that describes the task completed by the teleoperator. However, we find that these annotations often are imperfect along two key axes. First, datasets where a specific set of tasks is collected repeatedly at scale often have repetitive language instructions. For example, the dataset collected in Jang et al. (2022) contains 104 unique instructions for 39350 episodes, representing 0.26% unique instructions in the dataset. Secondly, as

Fig.4 · Architecture: VLM + per-layer KV → FM expert

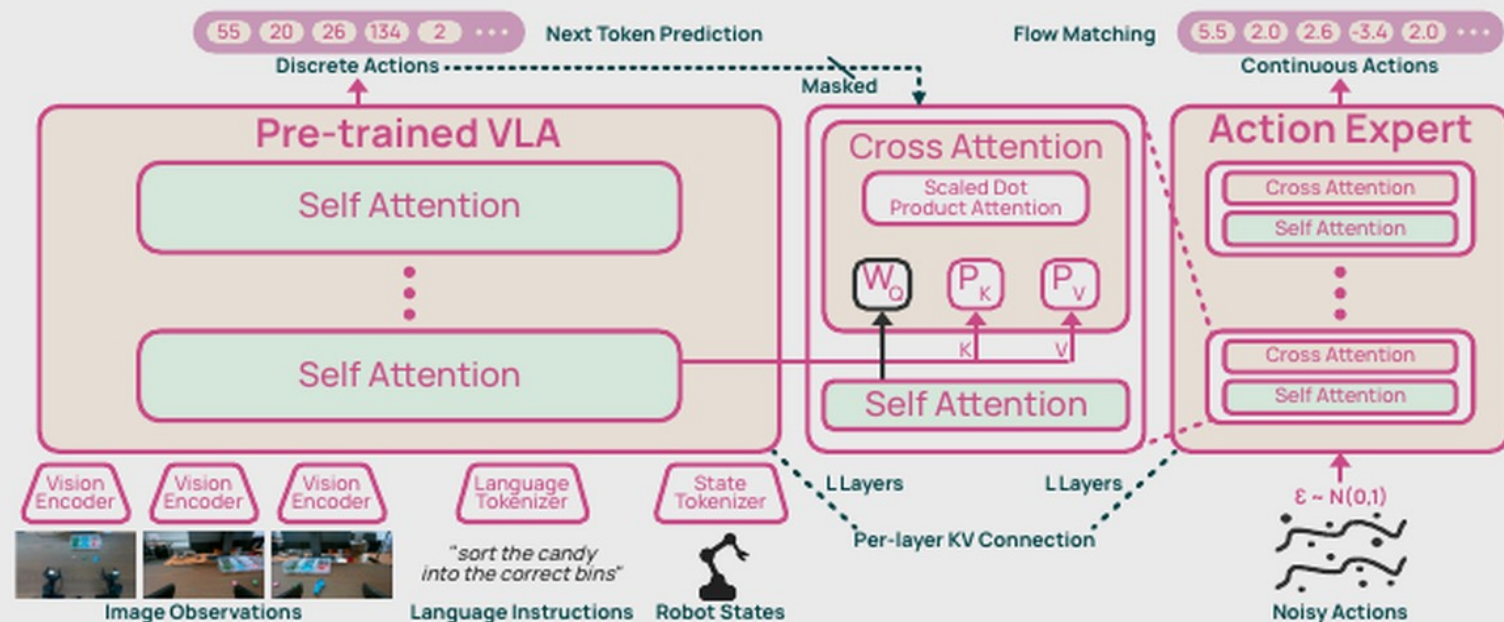


Figure 4 Overview of MolmoAct2. Image observations, language instructions, and robot states are tokenized and processed by a pre-trained VLA backbone with self-attention layers. Post-training attaches a DiT-style action expert with the same number of layers as the backbone. At each layer, the backbone key and value tensors are projected and reused as the key and value inputs to the action expert’s cross-attention, enabling per-layer KV conditioning from visual-language context to continuous control. The expert is trained with flow matching to denoise a noisy action trajectory into a continuous robot trajectory. During training, the backbone is also supervised with next-token prediction over discrete action tokens, while the target action-token span is masked from the expert so continuous action prediction cannot condition on the ground-truth discrete actions.

control strings using the same special-token wrappers used during training, e.g., `<setup_start>bimanual yam robotic arms in molmoact2<setup_end>`, and `<control_start>absolute joint pose<control_end>` or `<control_start>delta end-effector pose<control_end>`; the corresponding chat template and output trigger formatting are detailed in Appendix B.

Action and state representation. This stage is deliberately limited to discrete autoregressive supervision: the model predicts action tokens, but does not yet train the continuous action expert used in post-training. Each

Fig.7 · Real YAM 8-task FT

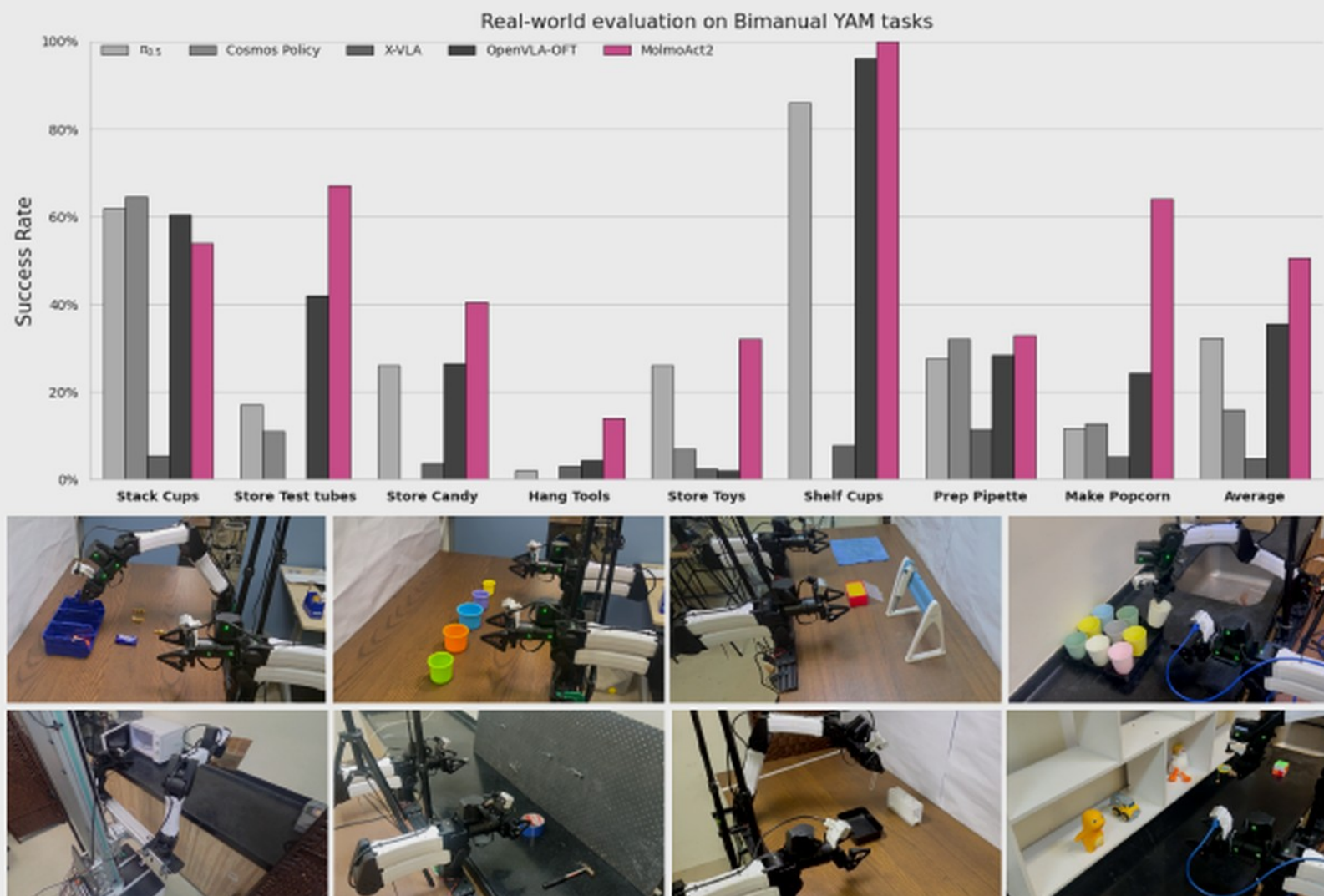


Figure 7 Overview of efficient fine-tuning of MolmoAct2. We conduct a comprehensive evaluation across 8 real-world tasks outside of a laboratory setting, ranging from preparing a pipette for a chemist to placing toys back onto a shelf. MOLMOACT2, fine-tuned on the collected evaluation data, outperforms 4 strong baselines by a large margin of 15% over the runner-up model.

Risks & limitations (honest)

Spatial OOD hard

Think OOD Spatial ~26% — geometry still brittle

Open articulations weak

MolmoSpaces Open 9.5% vs $\pi_{0.5}$ 22.7%

OOTB hardware bound

Verified mainly SO-100 & Franka DROID

Open-loop chunks

App E: no cross-chunk continuity; Hz $\neq$ reactivity

LIBERO ceilings

Spatial/Object ~98-100% $\rightarrow$ +0.9 avg hard to trust

Compute opacity

64×H100 pre/post not academic-default

Citation hygiene — FALSE / CLEAR / TENTATIVE

FALSE

“MolmoAct2 has $\pi_{0.5}$ HL subtasks”

Uses depth lattice, not semantic HL

FALSE

“Implements RTC”

RTC absent; cache + Graphs + Think reuse

FALSE

“Has BEHAVIOR System-2 voting”

Different intermediate (geometry $\neq$ stages)

CLEAR

Hybrid FAST $\rightarrow$ FM curriculum like $\pi_{0.5}$

Structural analogy; shared KI idea

CLEAR

Both call FM chunk expert (S1)

Different L1 tokens into same motor family

TENTATIVE

RTC wrap on Molmo FM

High-leverage follow-on — not in PDF

TENTATIVE

Depth + stage complementary

Likely for BEHAVIOR long-horizon

Stakeholder · close

Open ARM substrate when π is gated — competitive on paper protocols.

LIBERO 97.2/98.1 · Spaces 37.7 · DROID 87.1 · watch Spatial OOD + Open articulations

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MolmoAct2

Action Reasoning Models for Real-World Deployment

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🔥 **MolmoAct2:** MolmoAct2-Finetune MolmoAct2 MolmoAct2-Think MolmoAct2-Pretrain Molmo2-ER
📁 **MolmoAct2 Data:** MolmoAct2-BimanualYAM MolmoAct2-SO100/101 MolmoAct2-DROID Molmo2-ER
🔗 **Code:** [allenai/molmoact2](#)
💖 **Blog:** [allenai.org/blog/molmoact2](#)

Abstract



Vision-Language-Action (VLA) models aim to provide a single generalist controller for robots, but today’s systems fall short for real-world deployment. Frontier models are closed; open-weight alternatives are tied to expensive hardware; reasoning-augmented policies pay prohibitive latency for their grounding; and fine-tuned success rates remain below the threshold for dependable use. We present MOLMOACT2, a fully open action reasoning model built for practical deployment, advancing its predecessor, MOLMOACT along five axes. (1) MOLMOACT2 is built on top of our new MOLMO2-ER, a VLM backbone specialized for spatial and embodied reasoning, trained on a 3.3M-sample corpus with a specialize-then-rehearse recipe. (2) We release three new robot datasets spanning low-to-medium cost platforms: MOLMOACT2-BIMANUALYAM DATASET, 720 hours of teleoperated bimanual trajectories that constitute the largest open bimanual dataset to date; MOLMOACT2-DROID DATASET, a quality-filtered Franka subset of DROID; and MOLMOACT2-SO100/101 DATASET, a quality-filtered SO-100/101 subset. (3) We train and release MOLMOACT2-FAST TOKENIZER, an open-weight, open-data action tokenizer trained on millions of trajectories across five embodiments. (4) We design a new VLA architecture to graft the discrete-token VLM into the flow-matching continuous-action expert via per-layer key-value (KV) conditioning. (5) we propose MOLMOACT2-THINK, an adaptive-depth reasoning variant that re-predicts depth tokens only for scene regions that change between timesteps, retaining geometric grounding at a fraction of prior latency. In the most extensive empirical study of any open VLA to date, spanning 7 simulation and real-world benchmarks, MOLMOACT2 outperforms strong baselines including $\pi_{0.5}$, while MOLMO2-ER surpasses GPT-5 and Gemini Robotics ER-1.5 across 13 embodied-reasoning benchmarks. We release model weights, training code, and complete training data.